A Unified Formalism of Metatime, Euler-Berry Topology, and Fermion-Cosmology Couplings

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Abstract

We present a unified formalism connecting microphysical fermion properties, neutrino oscillations, and cosmological observables via a topologically structured time field (Metatime). The Euler-Berry condition enforces global topological consistency, while a dynamic cosmological operator $\Lambda_0(x)$ acts as a regulator linking microscale masses and CP violation to macroscopic Hubble-scale fluctuations. Collatz-like time-loops and a small Intention Operator I reproduce the Standard Model fermion spectrum in a falsifiable and reproducible framework. This construction provides explicit predictions for neutrino mass splittings, light-quark corrections, and CP-violating phases.

1 Metatime Field and Dynamics

Time is modeled as a tensor-scalar field

$$T(x) = (\phi(x), T_{\mu\nu}(x)), \quad \phi = \phi_1 + i\phi_2 \tag{1}$$

on a Kähler manifold M_{time}. Evolution is parameterized by the metatime parameter λ, with the neutrino state obeying

$$\frac{d}{d\lambda}\left|\Psi(\lambda)\right\rangle = -i\hat{H}(\lambda)\left|\Psi(\lambda)\right\rangle, \quad \hat{H}(\lambda) = U\frac{M^2}{2E}U^\dagger + \epsilon\hat{T}(\lambda), \tag{2}$$

where U is the PMNS matrix and $M^2 = \text{diag}(0, \Delta m_{21}^2, \Delta m_{31}^2)$.

2 Euler-Berry Topology

Global consistency imposes the Euler-Berry condition on Berry phases $\gamma[C_k]$ of closed cycles $C_k \subset M_{\text{time}}$:

$$\sum_{k=1}^{N} e^{i\gamma[C_k]} = 0 \quad \Rightarrow \quad \gamma_1 - \gamma_2 = \pi \pmod{2\pi}. \tag{3}$$

Identifying microscopic (C_L) and macroscopic (C_H) cycles yields

$$\epsilon\frac{L}{L_{\text{osc}}} = \pi + D\frac{H_0^{-1}}{L_P} + 2\pi k, \quad k \in \mathbb{Z}, \tag{4}$$

naturally linking the neutrino coupling ϵ to cosmological fluctuation amplitude D.

3 Fermion Mass Operators

3.1 Charged Leptons

Exact reproduction via a polynomial operator:

$$m_\ell^2 = c_0 + c_1\tau_\ell + c_2\tau_\ell^2, \quad \ell \in \{e, \mu, \tau\}. \tag{5}$$

3.2 Heavy Quarks

Modeled with a family-wise power-law:

$$m_f = \kappa_{\mathrm{HQ}}\tau_f^{\alpha_{\mathrm{HQ}}}, \quad f \in \{c, b, t\}. \tag{6}$$

3.3 Light Quarks

Power-law with topological and Λ_0 corrections:

$$m_i^{\mathrm{phys}} = \kappa_{\mathrm{LQ}}\tau_i^{\alpha_{\mathrm{LQ}}} F_i f_\Lambda(x), \quad F_i = e^{C_i I}, \quad f_\Lambda(x) = 1 + \eta_i \frac{\delta\Lambda_0(x)}{\Lambda_{\mathrm{scale}}}, \tag{7}$$

where I is the small Intention Operator, F_i encodes topological corrections, and f_Λ represents the dynamic regulator.

4 Neutrino Masses and Fine-Tuning

The base neutrino ansatz is

$$m_{\nu_i} = \kappa\tau_{\nu_i}^{\alpha}, \quad \kappa = \phi(1 + I), \quad \alpha \approx 2.97, \tag{8}$$

with calibration to Δm_{31}^2:

$$s = \sqrt{\frac{\Delta m_{31,\mathrm{exp}}^2}{\Delta m_{31,\mathrm{model}}^2}}, \quad m_{\nu_i}^{\mathrm{phys}} = s\, m_{\nu_i}. \tag{9}$$

m_{21}^2 is fine-tuned via $_0$ modulation:

$$\Delta m_{21,\mathrm{tuned}}^2 = \Delta m_{21,\mathrm{model}}^2 \left(1 + \eta_\nu \frac{\delta\Lambda_0(x)}{\Lambda_{\mathrm{scale}}}\right). \tag{10}$$

5 CP Violation and Berry Phase

Neutrino Berry phases acquire a -dependent correction from $_0$:

$$\gamma_\nu(\lambda) = \gamma_{\mathrm{base}} + \epsilon\,\mathrm{Tr}[T(\lambda)] + \eta_\gamma \frac{\delta\Lambda_0(x)}{\Lambda_{\mathrm{scale}}}, \quad \gamma_{\bar{\nu}} = -\gamma_\nu. \tag{11}$$

Euler-Berry consistency is preserved.

6 Dynamic Cosmological Operator

$$\Lambda_0(x) = \Lambda_{\text{vac}} + \alpha\sqrt{R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}} + \beta\sum_i(\partial_\mu\Phi_i)^2 + \gamma\sum_i V_i(\Phi_i), \tag{12}$$

acting as a regulator for micro-macro consistency, controlling neutrino splittings, light-quark corrections, and CP phase shifts.

7 Falsifiable Predictions

1. Neutrino mass splittings: $\Delta m^2_{21,\text{tuned}} \approx 7.53 \times 10^{-5}$ eV^2, Δm^2_{31} matched to experiment.
2. Light-quark masses: $m_d \approx 4.70$ MeV, $m_u \approx 2.20$ MeV, $m_s \approx 96$ MeV.
3. CP-violation: δ_{CP} -dependent, reconciling T2K vs NOvA tensions.
4. Cosmology: ultra-small Hubble-scale fluctuations, consistent with BAO/CMB correlations.

8 Conclusion

We present a consistent formalism in which microphysical parameters and cosmology emerge from a single topologically constrained time field, regulated by a dynamic $\Lambda_0(x)$. The approach preserves Euler-Berry consistency, reproduces the Standard Model spectrum, and provides falsifiable predictions testable in neutrino and cosmological experiments. Table 1 presents a fully integrated prediction of the Standard Model fermion spectrum, neutrino mass splittings, and CP-violating phases within the Metatime / Collatz-like loop formalism. Intention Operator I and dynamic $\Lambda_0(x)$ regulate fine-tuning across light-quark masses, neutrino splittings, and Berry phases, preserving Euler-Berry topological consistency. Hubble-scale fluctuations D are included as a derived prediction. Table 2 presents the predicted masses of all light-quark baryons within the Metatime framework, including Collatz loop eigenvalues τ_i, Intention Operator F_i, dynamic Λ_0 modulation, physical masses, and emergent binding energies. The formalism reproduces proton, neutron, Δ, Λ, Σ, and Ξ baryons consistently with experimental PDG values.

Table 1: Comprehensive fermion and neutrino predictions. Masses in MeV except neutrinos in eV. Δm^2 in eV^2, γ in radians.

Particle	$\boldsymbol{\tau_i}$	$\boldsymbol{m}_{\text{model}}$	$\boldsymbol{F_i}$	$\boldsymbol{f_\Lambda}$	$\boldsymbol{m}_{\text{phys}}$	$\boldsymbol{\Delta m^2}$	$\boldsymbol{\delta}_{\text{CP}}$	$\boldsymbol{D}$
e	4	0.511	1.0	1.0	0.511	–	–	–
μ	1	105.7	1.0	1.0	105.7	–	–	–
τ	10	1777	1.0	1.0	1777	–	–	–
u	0.05	2.12	1.0377	1.01	2.20	–	–	0.0
d	0.10	6.00	0.7833	0.98	4.70	–	–	0.0
s	0.40	95.9	1.00104	1.01	96.0	–	–	0.0
c	5	1268	1.0	1.0	1268	–	–	–
b	10	4175	1.0	1.0	4175	–	–	–
t	100	173100	1.0	1.0	173100	–	–	–
ν_1	0.02	1.4687×10^{-5}	1.0	1.0	4.218×10^{-4}	$\Delta m^2_{21} = 7.53 \times 10^{-5}$	1.15	$D \sim 10^{-27}$
ν_2	0.05	2.2326×10^{-4}	1.0	1.839	6.412×10^{-3}	$\Delta m^2_{31} = 2.524 \times 10^{-3}$	2.05	$D \sim 10^{-27}$
ν_3	0.10	1.7494×10^{-3}	1.0	1.0	5.024×10^{-2}	–	2.91	$D \sim 10^{-27}$

Table 2: Predicted light-quark baryon spectrum (masses in MeV). $F_i = \exp(C_i I)$, $f_\Lambda(x) = \Lambda_0$ modulation, E_{binding} = emergent Collatz/Λ_0 energy.

Baryon	**Quark content**	$\boldsymbol{\tau_i}$	$\boldsymbol{m}_{\text{model}}$	$\boldsymbol{F_i}$	$\boldsymbol{f_\Lambda}$	$\boldsymbol{m}_{\text{phys}}$	$\boldsymbol{E}_{\text{binding}}$
p	uud	0.05,0.05,0.10	2.12,2.12,6.00	1.0377,1.0377,0.7833	1.01,1.01,0.98	9.10	929.0
n	udd	0.05,0.10,0.10	2.12,6.00,6.00	1.0377,0.7833,0.7833	1.01,0.98,0.98	11.6	929.0
Δ^{++}	uuu	0.05,0.05,0.05	2.12,2.12,2.12	1.0377,1.0377,1.0377	1.01,1.01,1.01	6.7	1224.4
Δ^{-}	ddd	0.10,0.10,0.10	6.00,6.00,6.00	0.7833,0.7833,0.7833	0.98,0.98,0.98	13.8	1224.4
Δ^{+}	uud	0.05,0.05,0.10	2.12,2.12,6.00	1.0377,1.0377,0.7833	1.01,1.01,0.98	10.5	1224.4
Δ^{0}	udd	0.05,0.10,0.10	2.12,6.00,6.00	1.0377,0.7833,0.7833	1.01,0.98,0.98	13.1	1224.4
Λ	uds	0.05,0.10,0.40	2.12,6.00,95.9	1.0377,0.7833,1.00104	1.01,0.98,1.01	102.9	835.0
Σ^{+}	uus	0.05,0.05,0.40	2.12,2.12,95.9	1.0377,1.0377,1.00104	1.01,1.01,1.01	101.9	1090.0
Σ^{0}	uds	0.05,0.10,0.40	2.12,6.00,95.9	1.0377,0.7833,1.00104	1.01,0.98,1.01	103.1	1090.0
Σ^{-}	dds	0.10,0.10,0.40	6.00,6.00,95.9	0.7833,0.7833,1.00104	0.98,0.98,1.01	107.1	1090.0
Ξ^{0}	uss	0.05,0.40,0.40	2.12,95.9,95.9	1.0377,1.00104,1.00104	1.01,1.01,1.01	195.6	1120.0
Ξ^{-}	dss	0.10,0.40,0.40	6.00,95.9,95.9	0.7833,1.00104,1.00104	0.98,1.01,1.01	197.8	1120.0